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Roll No................

EVEN END SEMESTER EXAMINATION – 2024

Name of the Course: B-Tech (ME)

Semester: VI

Name of the Paper: Automotive Mechatronics

Paper Code: TMS-601

Time: 03 Hours

Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer any tow sub questions among a, b, & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	Describe a typical mechatronic system used in automobiles and its components. Provide an example of such a system.	**CO1**
(b)	Explain the importance of the body subsystem in an automobile. Discuss its main functions and materials used in its construction.	**CO1**
(c)	Explain the classification of automobiles based on their fuel types. Discuss the advantages and disadvantages of each fuel type.	**CO1**
Q2	(20 marks)	
(a)	Define an electric motor and discuss its significance in automobile applications. Explain the types of electric motors used in vehicles and their respective functions.	**CO2**
(b)	Discuss the importance of instrumentation in modern automobiles. Explain the various types of sensors and instruments used to monitor vehicle performance and conditions.	**CO2**
(c)	Explain the function of instrumentation gauges in an automobile. Discuss the different types of gauges commonly found on a car's dashboard and their respective purposes.	**CO2**
Q3	(20 marks)	

(a)	Define an automotive sensor and explain its importance in vehicle systems. Discuss the various categories of sensors commonly used in modern automobiles and their general working principles.	**CO3**
(b)	Explain the role of a crankshaft position sensor in an automobile engine. Discuss its working principle and the importance of accurate crankshaft position data for engine performance.	**CO3**
(c)	Discuss the function of a speed sensor in an automobile. Explain its working principle and the types of speed sensors commonly used in vehicles.	**CO3**
Q4	(20 marks)	
(a)	Compare FlexRay and MOST communication protocols in terms of speed, and application scenarios. Discuss their roles in enhancing vehicle performance and entertainment systems, respectively.	**CO4**
(b)	Define the Internet of Things (IoT) and explain its relevance to the automotive industry. Discuss how IoT technologies can transform the driving experience and vehicle functionality.	**CO5**
(c)	List and briefly describe the main communication protocols used in modern automobiles. Discuss the differences between these protocols in terms of speed, reliability, and application.	**CO4**
Q5	(20 marks)	
(a)	Discuss the benefits and potential challenges associated with implementing IoT solutions in vehicles.	**CO5**
(b)	Define a hybrid vehicle and explain its operating principle. Discuss the advantages and challenges of hybrid technology in the context of modern transportation.	**CO6**
(c)	Describe the potential societal impacts of autonomous vehicles, including safety improvements, traffic management, and changes in transportation habits. Discuss the ethical considerations associated with autonomous driving.	**CO6**

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Course: B.Tech (ME)
Name of the Paper: Design of Machine Elements II

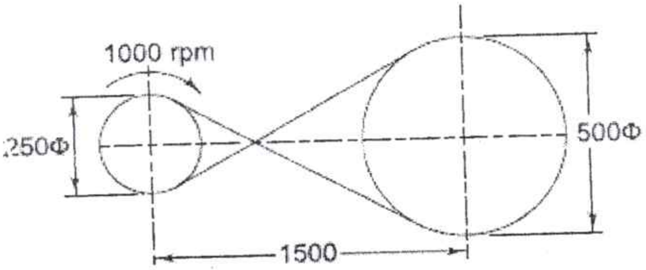
Semester: VI
Paper Code: TME 602

Time: 3 Hours

Maximum Marks: 100

Note:

(i) All questions are compulsory
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each question are twenty
(iv) Each question carries 10 marks
(v) Use of Design Data Handbook and Calculator (Non-Programmable) is permissible

Q1	(20 marks)	
a)	Classify springs according to their shapes. Draw neat sketches indicating in each case whether stresses are induced by bending or by torsion. Discuss the design analysis for a helical compression spring.	CO1
b)	A semi-elliptic leaf spring consists of two extra full-length leaves and six graduated - length leaves, including the master leaf. Each leaf is 7.5 mm thick and 50 mm wide. The centre-to-centre distance between the two eyes is 1 m. The leaves are pre-stressed in such a way that when the load is maximum, stresses induced in all the leaves are equal to 350 N/mm^2. Determine the maximum force that the spring can withstand.	
c)	What is nipping in a leaf spring? Discuss its role. List the materials commonly used for the manufacture of the leaf springs? What do you understand by full length and graduated leaves of a leaf spring? Write the expression for determining the stress and deflection in full length and graduated leaves.	
Q2	(20 marks)	
a)	The layout of a crossed leather belt drive transmitting 7.5 kW is shown in Fig. 1. The mass of the belt is 0.55 kg per metre length and the coeffi -cient of friction is 0.30. Calculate (i) the belt tensions on the tight and loose sides, and (ii) the length of the belt.  Fig. 1	CO2
b)	A flat belt, 8 mm thick and 100 mm wide transmits power between two pulleys, running at 1600 m/min. The mass of the belt is 0.9 kg/m length. The angle of lap in the smaller pulley is 165° and the coefficient of friction between the belt and pulleys is 0.3. If the maximum permissible stress in the belt is 2 MN/m2, find (i) Maximum power transmitted, and (ii) Initial tension in the belt.	

MN/m^2

c)	Sketch the cross section of a V-belt and label its important parts with brief explanation. How will you designate V-belt? State advantages and disadvantages of V-belt drive over flat belt drive.	
Q3	(20 marks)	CO3
a)	How are the gears classified and what are the various terms used in spur gear terminology? Discuss the design procedure of spur gears.	
b)	A pair of parallel helical gears consists of an 18 teeth pinion meshing with a 45 teeth gear.7.5 kW power at 2000 rpm is supplied to the pinion through its shaft. The normal module is 6 mm, while the normal pressure angle is 20°. The helix angle is 23°. Determine the tangential, radial and axial components of the resultant tooth force between the meshing teeth.	
c)	State the two most important reasons for adopting involute curves for a gear tooth profile. Explain the phenomenon of interference in involute gears. What are the conditions to be satisfied in order to avoid interference ?	
Q4	(20 marks)	CO4
a)	A pair of worm gears is designated as 1/52/10/8. The worm rotates at 1000 rpm and the normal pressure angle is 20°. Determine the coefficient of friction and the efficiency of the worm gears.	
b)	How the bevel gears are classified? Explain with neat sketches. What are the various forces acting on a bevel gear and on worm and worm gears ?	
c)	A pair of bevel gears consists of a 30 teeth pinion meshing with a 48 teeth gear. The gears are mounted on shafts, which are intersecting at right angles. The module at the large end of the tooth is 4 mm. Calculate (i) the pitch circle diameters of the pinion and the gear; (ii) the pitch angles for the pinion and gear; and (iii) the cone distance.	
Q5	(20 marks)	CO5/CO6
a)	A ball bearing is subjected to a radial force of 2500 N and an axial force of 1000 N. The dynamic load carrying capacity of the bearing is 7350 N. The shaft is rotating at 720 rpm. Calculate the life of the bearing.	
b)	(i) Enumerate applications of sliding-contact and rolling-contact bearings. (ii) Define- L_{10} life, Dynamic bearing Capacity, Equivalent bearing load, Bearing Modulus, Sommerfeld's number	
c)	A 360° hydrodynamic bearing operates under the following conditions: radial load = 50 kN journal diameter = 150 mm bearing length = 150 mm radial clearance = 0.15 mm minimum fi lm thickness = 0.03 mm viscosity of lubricant = 8 cP What is the minimum speed of operation for the journal to work under hydrodynamic conditions?	

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END SEMESTER EXAMINATION JUN/JUL - 2024

Name of the Course: _BTECH ME Semester: _VI
Name of the Paper: Industrial Engineering and Project Management
: _**TME 609**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks
(v) Answers must be accompanied by neat and labelled diagrams
(vi) Suitable assumptions can be made for any missing data

Q1	(20marks)	
(a)	Critically examine the definition of productivity in the context of a manufacturing company. How does this definition impact the evaluation of employee performance and company success?	CO1
(b)	Analyze how technological advancements and employee training programs affect productivity in the tech industry. Discuss the relationship between these factors and overall business performance.	CO1
(c)	Discuss how the consideration of human factors in the work study process can lead to improvements in job design and work flow. Use examples from the manufacturing sector.	CO1
Q2		
(a)	Create a detailed proposal for redesigning a warehouse operation that employs work simplification and ergonomic principles to enhance worker safety and operational efficiency. Include diagrams and a step-by-step process for the implementation of your design.	CO2
(b)	Apply the concept of rating factors to establish the standard time for assembling a consumer electronics product. Outline the process and considerations necessary to determine accurate standard times.	CO2
(c)	Develop a comprehensive plan to implement advanced time study techniques in a large manufacturing plant. Your plan should include steps for data collection, analysis, and application of results to improve throughput times. Discuss potential challenges and how they could be addressed.	CO2
Q3	(20marks)	
(a)	Explain the concept of Material Resource Planning as applicable to a manufacturing firm.	CO3
(b)	Give the details of the seven quality tools with relevant manufacturing examples.	CO3
(c)	Justify the use of 5s in the manufacturing area . Explain how lean manufacturing techniques can improve the profitability of a company.	CO3
Q4	(20marks)	
(a)	The following data gives the automobile sales of the company for various years. Fit the straight line. Forecast the sales for the year 2024 and 2025.	C04

	<table><tr><td>Year</td><td>Sales (000)</td></tr><tr><td>2015</td><td>13</td></tr><tr><td>2016</td><td>20</td></tr><tr><td>2017</td><td>20</td></tr><tr><td>2018</td><td>28</td></tr><tr><td>2019</td><td>30</td></tr><tr><td>2020</td><td>32</td></tr><tr><td>2021</td><td>33</td></tr><tr><td>2022</td><td>38</td></tr><tr><td>2023</td><td>43</td></tr></table>	
(b)	Give a detailed account of the costs involved in inventory management. Explain how JIT affects inventory management.	C04
(c)	ABC Corporation has got a demand for particular part at 10,000 units per year. The cost per unit is ₹ 7 and it costs ₹ 30 to place an order and to process the delivery. The inventory carrying cost is estimated at 9 per cent of average inventory investment. Determine (i) Economic order quantity. (ii) Optimum number of orders to be placed per annum. (iii) Minimum total cost of inventory per annum.	CO4
Q5	(20marks)	
(a)	The following details are available regarding a project: <table><tr><td>Activity</td><td>Predecessor Activity</td><td>Duration (Weeks)</td></tr><tr><td>A</td><td>-</td><td>3</td></tr><tr><td>B</td><td>A</td><td>5</td></tr><tr><td>C</td><td>A</td><td>7</td></tr><tr><td>D</td><td>B</td><td>10</td></tr></table>	C05

E	C	5
F	D,E	4

Determine the critical path, the critical activities and the project completion time.

(b)	Explain the concepts involved PERT and its applicability to projects in Engineering and compare PERT and CPM.	CO5
(c)	With a relevant example, explain the activities involved in the life cycle of a project.	CO5

~~Note for the question paper setters:~~

- ~~**Question paper should cover all the COs of the course.**~~

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End Semester Examination 2024

Name of the Course: B.Tech — Semester: VI

Name of the Paper– Industrial Automation

PaperCode: TOE 603

Time: 3Hours — Maximum Marks: 100

Note: -

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Attempt any FIVE of the following. i) State the benefit of Automation. ii) Compare Fixed and Modular PLC. iii) State the I/O Module selection criteria with respect to PLC. iv) Give list of any four relay type instructions with their symbols. v) List the functions of Electrical drives. vi) List different editors used in SCADA.	**CO1, CO2**
(b)	Attempt any Two of the following. i) Compare fixed and flexible automation on any four points. ii) Explain redundancy in PLC. iii) Draw a neat block diagram of PLC and describe the working of its parts.	
(c)	Attempt any Two of the following. i) List any four device & four output devices that can be connected to PLC. ii) Compare PLC and SCADA system on any four points. iii) Write a PLC ladder program for start stop logic with two normal open switch	
Q2		**CO1,CO2**
(a)	Explain the sinking & sourcing concept in PLC input output module.	
(b)	Describe the working of UP counter with neat diagram and waveform	
(c)	Draw and explain the ON delay timer instruction. State the function of following : i) Enable bit (EN) ii) Done bit (DN) iii) Timer timing bit (TT)	
Q3		**CO3**
(a)	Describe the steps involve developing SCADA application for following system.	
(b)	Draw Ladder diagram for automatic bottle filling system. Assume suitable system design for the same.	
(c)	List different PLC programming languages. Explain any one with example.	
Q4		**CO4**
(a)	Explain the concept of a Distributed Control System (DCS) and discuss its role in industrial automation. Compare and contrast DCS with other control systems such as PLCs and SCADA.	
(b)	Explain the role of advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML) in Distributed Control Systems (DCS). How can AI/ML algorithms enhance predictive maintenance, anomaly detection, and process optimization.	
(c)	Explain the cyber security challenges associated with Distributed Control Systems (DCS). How can DCS architectures be hardened against cyber threats?	

Q5		CO5
(a)	What are some common challenges associated with deploying and maintaining RTUs in industrial environments?	
(b)	What is a Remote Terminal Unit (RTU) and what role does it play in industrial automation?	
(c)	Define Phasor Measurement Units (PMUs) and explain their significance in modern power systems. Discuss the key parameters measured by PMUs.	

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END SEMESTER EXAMINATION - 2024

Name of the Course: **B.Tech**
Name of the Paper: **Computer Aided Manufacturing**

Semester: **VI**
Paper Code: **TME 606**

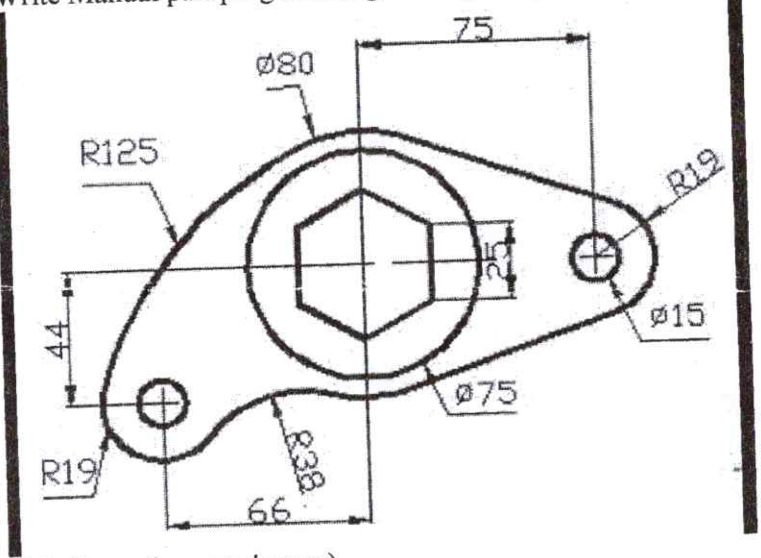
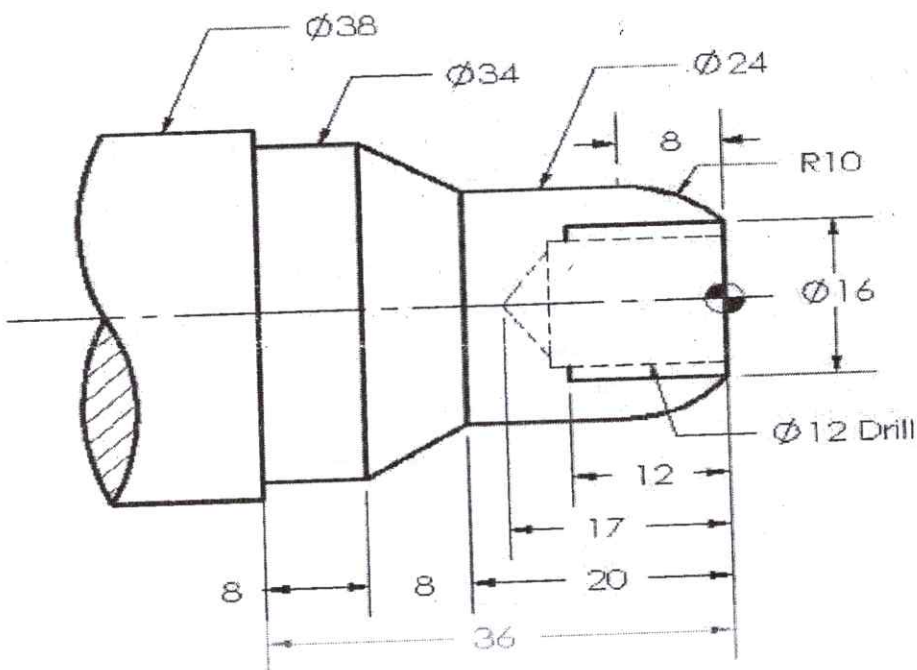
Time: **3 Hours**

Maximum Marks: **100**

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20marks)	CO1, CO2, CO3
(a)	Compare and contrast the social and environmental consequences of the Industrial Revolution with those of Industry 4.0, considering issues such as urbanization, resource consumption, and sustainability practices.	
(b)	i. Give the general configuration of a CAD computer system. ii. In what ways CAD can help manufacturing activity? Discuss. iii. CAD helps in integrating CAM- Justify this statement.	
(c)	Explore the integration of automation and robotics with NC machine tools in modern manufacturing facilities. What are the advantages of incorporating automation into NC machining processes, and how does this integration affect productivity, flexibility, and overall manufacturing efficiency?	
Q2	(20 marks)	CO2, CO3
(a)	Explain how CNC and DNC technologies contribute to the optimization and automation of manufacturing processes. What role do machine learning play in enhancing the performance of CNC machines and DNC networks?	
(b)	Discuss the advantages and disadvantages of a star network topology compared to a bus network topology. How does each topology affect factors such as scalability, fault tolerance, and ease of maintenance?	
(c)	Explore the impact of NC (Numerical Control) machine tools on the evolution of modern manufacturing. What specific advantages do NC machines provide in terms of precision, repeatability, and complex part production, and how do these benefits influence overall manufacturing strategies? Conversely, what are the key limitations or risks associated with relying solely on NC machining processes?	
Q3	(20 marks)	CO1, CO2,
(a)	Define the term "Group Technology" and its importance in plant layout and manufacturing process planning .	
(b)	Briefly describe the features of F.M.S .	
(c)	i. Discuss the difficulties encountered in using conventional numerical control. ii. Enumerate the advantages of Computer Assisted Part Programming when compared to Manual Part Programming.	
Q4	(20 marks)	CO2, CO3, CO4
(a)	i. Explain the G codes used for tool offset functions? ii. Describe the features of CNC machining centers.	
(b)	How has automation transformed the landscape of Computer-Aided Manufacturing (CAM) in recent years, and what are the key technological advancements driving this transformation? Discuss the implications of automation on efficiency, precision, and scalability within CAM processes.	
(c)	Compare and contrast open-loop and closed-loop control systems in terms of their operational principles, applications, advantages, and limitations. How does feedback	

	play a role in closed-loop systems, and what are the implications of lacking feedback in open-loop systems?	
Q5	(20 marks)	CO2, CO5, CO6
(a)	What are the three major elements of an ASRS? Explain.	
(b)	Write Manual part programming of the given geometry.  (All dimensions are in mm)	
(c)	Write Manual part programming of the given geometry.  (All dimensions are in mm)	

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EVENEND SEMESTER EXAMINATION - 2024

Name of the Course: **B. Tech**Semester: **VI**th
Name of the paper: **Pressure Die Casting**Paper Code:**DEMM11**
Time: **3 hours** Maximum Marks: **100**

Note: -

(i) All questions are compulsory
(ii) Answer any two sub questionsamong a, b, & c in each main question
(iii) Total marks in each main question are twenty
(iv) Each question carries 10 marks

Q1.	**20 marks**	
(a)	Define High pressure Die casting (HPDC) and explain its advantages over traditional casting methods. Discuss specific equipment used in HPDC and outline a maintenance practice to ensure its optimal performance.	**CO1**
(b)	Identify casting defect related to shrinkage and explain its causes. Propose remedy to address this defect, considering both process adjustments and material modifications.	
(c)	Discuss the purpose of heat treatment in casting processes and its impact on the mechanical properties of casted components.	
Q2.	**20 marks**	
(a)	Describe the principle behind vacuum casting and its advantages over conventional casting methods. Briefly explain one application where vacuum casting is particularly beneficial.	**CO2**
(b)	Define alloying elements and provide two examples of commonly used alloying elements in metal alloys. Briefly explain one desirable property imparted by each of these alloying elements to the alloy.	
(c)	Provide an overview of the various methods encompassed by surface heat treatment, including their general objectives and applications.	
Q3.	**20 marks**	
(a)	Describe the components of a typical gating system, including the sprue, runner, and gates, and their functions in controlling the metal flow.	**CO3**
(b)	Describe the important methods of Riser design in gating system.	
(c)	A cylindrical job with diameter of 200 mm and height of 100 mm is to be cast using modulus method of riser design. Assume that the bottom surface of cylindrical riser does not contribute as cooling surface. If the diameter of the riser is equal to its height, then the height of the riser in mm?	
Q4.	**20 marks**	
(a)	A Spherical drop of molten metals of radius 5 mm was found to solidify in 15 seconds. A similar drop of radius 9 mm would solidify in how many seconds?	**CO4**
(b)	Explain the function of Riser. Why it is necessary to design proper riser. What	

	guidelines to be follow for designing the riser?	
(c)	The length of a mold sprue is 55 cm and the cross-sectional area at its base is 6 cm^2. The sprue feeds a horizontal runner leading into a mold cavity whose volume is 2700 cm^3. Determine the time to fill the mould cavity in seconds. Assume g= 10 m/sec^2.	
Q5.	**20 marks**	
(a)	Define die maintenance and explain its importance in the die casting industry. Provide two examples of common maintenance practices performed on dies and their impact on production efficiency.	**CO5**
(b)	Discuss the significance of coatings for dies in die casting operations. Name two types of coatings commonly used for dies and explain their specific properties and applications.	
(c)	Define Industry 4.0 and outline its key concepts related to automation, robotics, artificial intelligence, and the industrial internet of things (IIoT). Provide one example of how each concept is transforming the die casting industry.	